OFFROAD TERRAIN SEMANTIC SEGMENTATION

FUNDAMENTAL IN AIML PROJECT

2025-2026

SUBMITTED BY-

NAME-ANUSHKA YADAV

REGISTRATION NO.-25BCE10978

1. ABSTRACT

Autonomous systems operating in off-road environments require robust perception capabilities to understand complex and unstructured terrains. Unlike structured urban roads, off-road scenes contain irregular surfaces, vegetation, rocks, and water bodies, making segmentation a challenging task.

This project presents a deep learning-based approach for **semantic segmentation of off-road terrain** using a **UNet architecture with a MobileNetV2 encoder**. The model is trained on a synthetic dataset consisting of annotated terrain images and corresponding segmentation masks.

The trained model achieves a **pixel accuracy of 84.58%** and a **mean Intersection over Union (mIoU) of 46.04%**, demonstrating its ability to effectively classify terrain categories. The training and validation loss curves show consistent convergence, indicating stable learning and minimal overfitting. This system can serve as a

foundational perception module for autonomous vehicles and robotics applications.

2. INTRODUCTION

Semantic segmentation plays a critical role in computer vision by enabling pixel-level understanding of scenes. In off-road environments, this task becomes significantly more challenging due to:

- Absence of well-defined roads
- Presence of diverse terrain types
- High variability in lighting and textures

Accurate segmentation allows autonomous systems to:

- Identify navigable paths
- Avoid obstacles
- Adapt to environmental changes

This project focuses on building a reliable segmentation model capable of distinguishing multiple terrain types in off-road scenarios.

3. PROBLEM STATEMENT

The objective of this project is to design and implement a deep learning model that can:

- Perform **pixel-wise classification** of off-road terrain images
- Identify multiple terrain categories accurately
- Maintain **generalization** across unseen validation data

4. OBJECTIVES

- Develop a semantic segmentation pipeline using deep learning
- Train a UNet-based model for terrain classification
- Evaluate model performance using standard metrics
- Generate visual outputs for better interpretability

5. DATASET DESCRIPTION

The dataset used in this project consists of synthetic off-road terrain images.

Key Details:

- **Training Images:** 2,857
- **Validation Images:** 317
- **Input Type:** RGB images
- **Annotation Type:** Grayscale masks

Classes:

Class ID	Description
0	Sky / Background
1	Ground Terrain
2	Vegetation / Shrubs
3	Rocks / Obstacles
4	Water / Flat Surfaces
5	Other Terrain

Each pixel in the mask corresponds to a specific terrain class, enabling supervised learning.

6. METHODOLOGY

6.1 Data Preprocessing

To ensure uniformity and efficient training:

- Images resized to **256×256 pixels**
- Pixel values normalized using ImageNet statistics
- Masks converted from raw pixel values to class indices

6.2 Data Augmentation

To improve generalization and reduce overfitting:

- Horizontal flipping (random)
- Brightness and contrast adjustment

- Small rotations

These transformations simulate real-world variations.

6.3 Model Architecture

The model is based on the **UNet architecture**, which consists of:

- **Encoder (MobileNetV2):** Extracts hierarchical features
- **Decoder:** Reconstructs spatial details for segmentation
- **Skip connections:** Preserve fine-grained information

Why MobileNetV2?

- Lightweight and efficient
- Faster training
- Suitable for limited computational resources

6.4 Training Configuration

Parameter	Value
Loss Function	CrossEntropyLoss
Optimizer	AdamW
Learning Rate	0.0001
Batch Size	8
Epochs	5
Scheduler	ReduceLROnPlateau

6.5 Training Process

The training pipeline follows these steps:

1. Load batch of images and masks
2. Forward pass through the model

3. Compute loss using predicted and ground truth masks
4. Backpropagation to update weights
5. Validate model after each epoch

The best model is saved based on **lowest validation loss**.

7. RESULTS

7.1 Quantitative Performance

Metric	Value
Pixel Accuracy	84.58%
Mean IoU	46.04%
Validation Loss	0.4049

7.2 Loss Curve Analysis

- **Training Loss:** Decreased from ~0.87 to ~0.45
- **Validation Loss:** Decreased from ~0.53 to ~0.40

Observations:

- Continuous decrease in both losses indicates effective learning
- No sharp divergence → **low overfitting**
- Validation loss closely follows training loss → **good generalization**

7.3 Qualitative Results

The model produces:

- Segmentation masks
- Color-coded outputs
- Overlay images combining predictions and original inputs

These outputs visually confirm that the model can:

- Identify terrain regions
- Distinguish obstacles and vegetation
- Highlight navigable paths

8. EVALUATION METRICS

8.1 Pixel Accuracy

Measures overall correctness:

$$Accuracy = \frac{Correct\ Pixels}{Total\ Pixels}$$

8.2 Intersection over Union (IoU)

Evaluates overlap between prediction and ground truth:

$$IoU = \frac{Intersection}{Union}$$

8.3 Mean IoU

Average IoU across all classes:

$$mIoU = \frac{1}{N}\sum IoU_i$$

9. SYSTEM IMPLEMENTATION

Technologies Used:

- **Programming Language:** Python
- **Framework:** PyTorch
- **Libraries:**
  - segmentation-models-pytorch
  - OpenCV
  - Albumentations

- NumPy
- Matplotlib

Project Modules:

- **train.py:** Handles model training and loss plotting
- **evaluate.py:** Computes accuracy and IoU metrics
- **test.py:** Generates predictions and visualizations

10. CHALLENGES FACED

- Handling different mask formats and pixel mappings
- Managing class imbalance in terrain types
- Limited training epochs due to time constraints
- Ensuring correct pairing of images and masks

11. LIMITATIONS

- Model trained for only 5 epochs
- Moderate IoU suggests room for improvement
- Synthetic dataset may not fully capture real-world complexity

12. FUTURE ENHANCEMENTS

- Increase training duration and dataset size
- Use advanced architectures (DeepLabV3+, EfficientNet)
- Implement hybrid loss functions (Dice + CrossEntropy)
- Deploy model for real-time inference
- Use real-world off-road datasets

13. CONCLUSION

This project successfully demonstrates the application of deep learning for semantic segmentation of off-road terrain. The UNet model with MobileNetV2 encoder provides a balance between efficiency and performance.

The results show that the model is capable of learning meaningful representations of terrain features and can generalize reasonably well. With further improvements, this system can be extended for real-world autonomous navigation applications.

14. REFERENCES

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3. segmentation-models-pytorch Documentation

4. Albumentations Library Documentation